Project – 13
Batch - 14

Road Accident Data Analysis System

A web-based application that displays road accident statistics by year and location to identify accident trends and improve road safety awareness.

Team Contributions

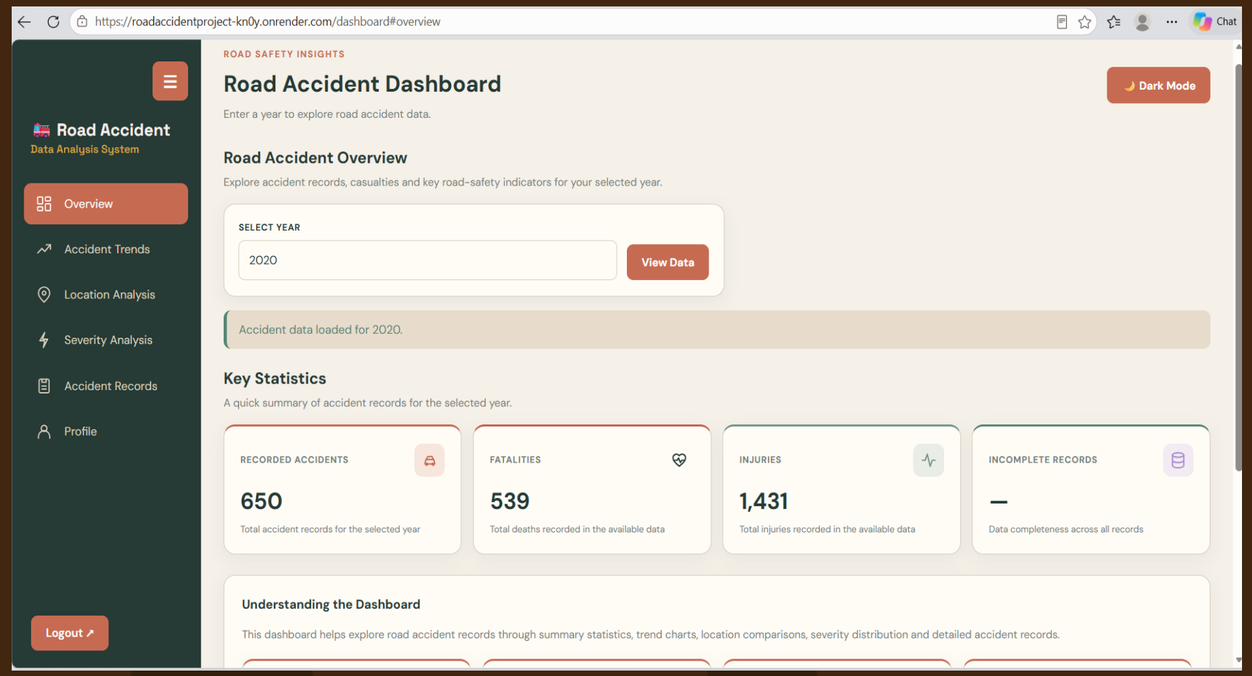
Sri Hari - Database Analyst

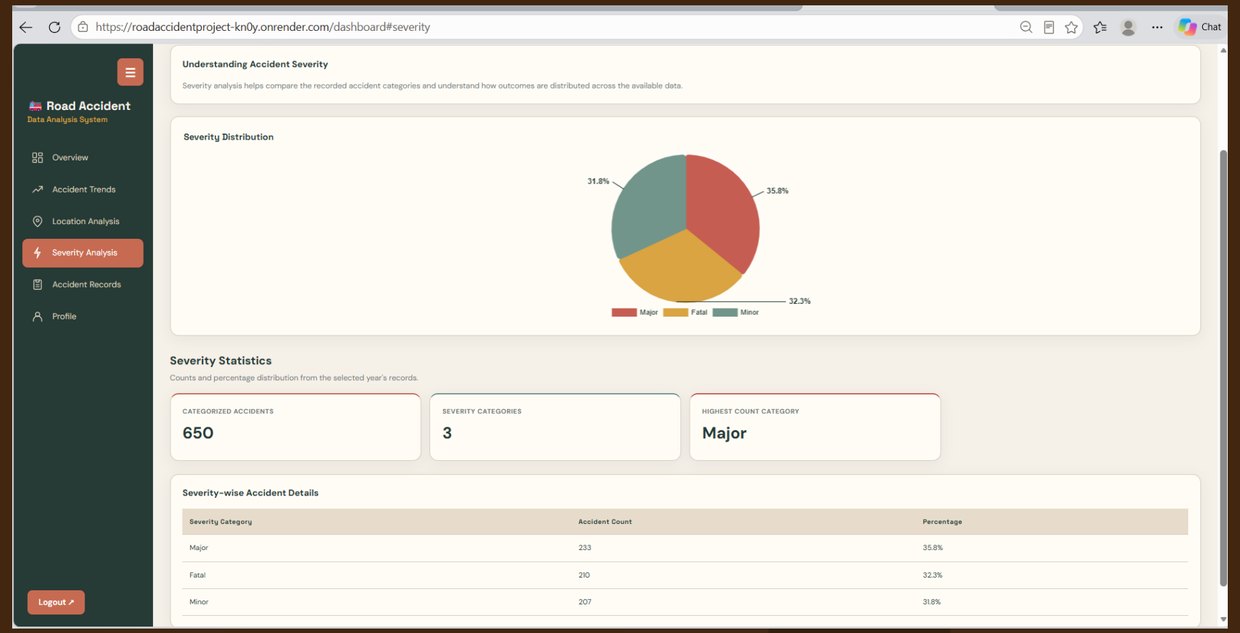
Gouthami - Database Developer

Venkata Charan - Backend Developer

Harsha Bindu - Frontend Developer

Data analysis and visualization (Sri Hari)






```
12  SELECT * FROM accidents LIMIT 10;
```

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Output | Result #7

accident_id	accident_date	accident_time	location	location_id	vehicle_type	severity	deaths	injuries	weather	road_name
16001	05/06/2011	7:37:00 AM	Chandragiri Road	57	Auto	Minor	0	1	Sunny	Chandragiri Road
16002	02/17/2011	7:27:00 AM	Bengaluru Main Road	58	Auto	Minor	0	1	Cloudy	Bengaluru Main Road
16003	10/29/2011	9:29:00 AM	Outer Ring Road	59	Truck	Minor	0	1	Rainy	Outer Ring Road Road
16004	06/22/2011	3:29:00 AM	Chennai Main Road	60	Bike	Minor	0	2	Rainy	Chennai Main Road
16005	01/23/2011	3:40:00 PM	NH 44	61	Bike	Fatal	1	3	Windy	NH 44
16006	10/23/2011	6:33:00 AM	NH 44	61	Car	Fatal	1	0	Rainy	NH 44
16007	07/14/2011	9:29:00 AM	Chittoor Bypass	62	Bus	Major	1	2	Rainy	Chittoor Bypass Road
16008	11/08/2011	9:40:00 PM	Khammam Bypass	63	Car	Minor	0	1	Foggy	Khammam Bypass Road
16009	12/17/2011	11:04:00 AM	Mysuru Road	64	Bike	Minor	0	1	Rainy	Mysuru Road Road
16010	04/19/2011	7:21:00 PM	Mysuru Road	64	Truck	Fatal	2	1	Foggy	Mysuru Road Road

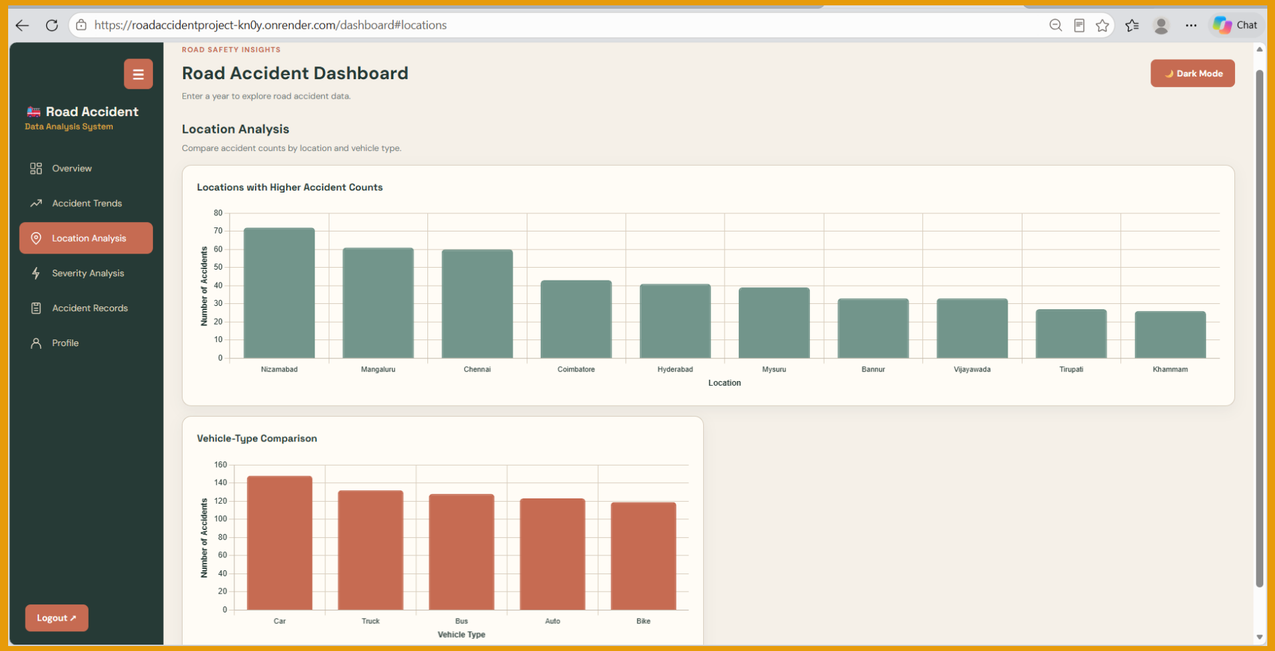


Figure : Location Analysis: Accident Counts by Location and Vehicle Type

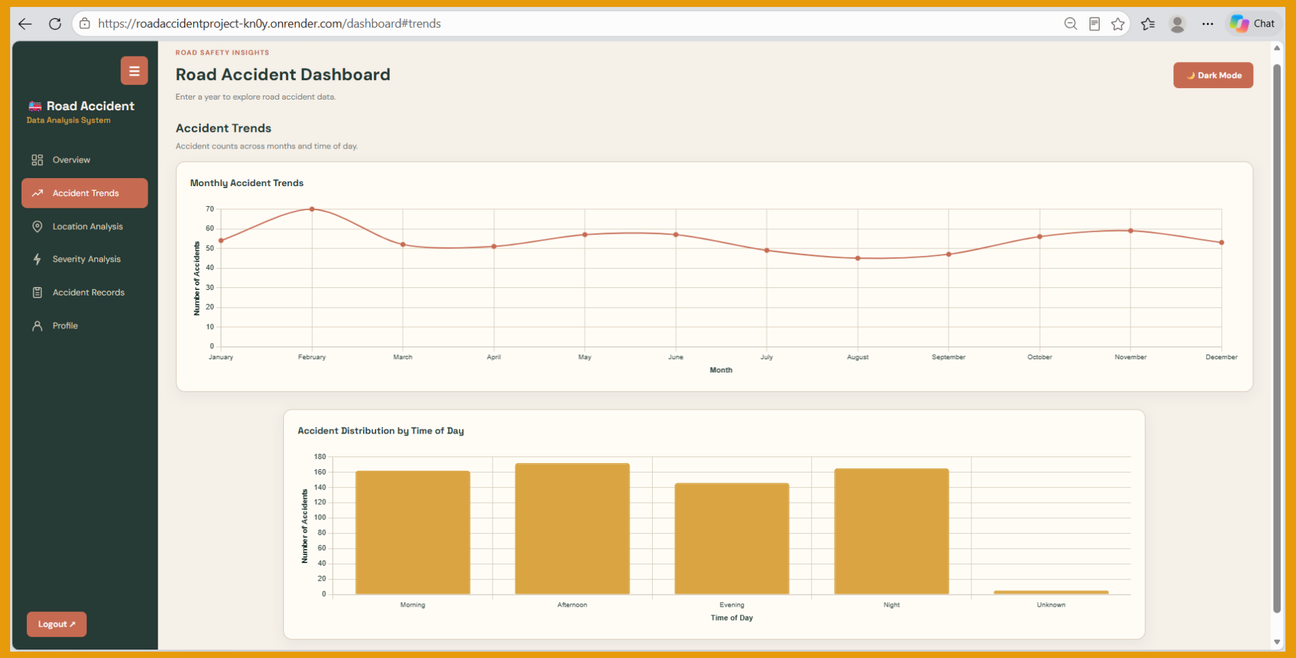


Figure : Accident Trends: Monthly Trends and Time-of-Day Distribution

Contributer – Munnam Sri Hari Vardhan Reddy

Role – Data analyst and visualization developer

Technologies used:

Python,
Pandas,
Matplotlib

Implementation:

1. Explored and organized road accident datasets.
2. Analyzed accident statistics and trends.
3. Prepared data for charts and visualizations.

Outcome:

Organized accident data for analysis and helped identify accident patterns and trends.

Database design and management (Gowthami)

Output Result #4

Field	Type	Null	Key	Default	Extra
accident_id	int	NO	PRI	NULL	auto_increment
accident_date	date	NO		NULL	
accident_time	time	YES		NULL	
location	varchar(255)	YES		NULL	
location_id	int	YES	MUL	NULL	
vehicle_type	varchar(50)	YES		NULL	
severity	varchar(30)	YES		NULL	
deaths	int	YES		0	
injuries	int	YES		0	
weather	varchar(50)	YES		NULL	
road_name	varchar(255)	YES		NULL	

Figure: Database tables and SQL query output

```
7   •USE road_accident_db;
8
9   •SELECT * FROM accidents
10   LIMIT 10;
11
12
```

Ln

Output Result #6

accident_id	accident_date	accident_time	location	location_id	vehicle_type	severity	deaths	injuries	weather	road_name
6001	05/06/2011	7:37:00 AM	Chandragiri Road	57	Auto	Minor	0	1	Sunny	Chandragiri Road
6002	02/17/2011	7:27:00 AM	Bengaluru Main Road	58	Auto	Minor	0	1	Cloudy	Bengaluru Main Road
6003	10/29/2011	9:29:00 AM	Outer Ring Road	59	Truck	Minor	0	1	Rainy	Outer Ring Road Road
6004	06/22/2011	3:29:00 AM	Chennai Main Road	60	Bike	Minor	0	2	Rainy	Chennai Main Road
6005	01/23/2011	3:40:00 PM	NH 44	61	Bike	Fatal	1	3	Windy	NH 44
6006	10/23/2011	6:33:00 AM	NH 44	61	Car	Fatal	1	0	Rainy	NH 44
6007	07/14/2011	9:29:00 AM	Chittoor Bypass	62	Bus	Major	1	2	Rainy	Chittoor Bypass Road
6008	11/08/2011	9:40:00 PM	Khammam Bypass	63	Car	Minor	0	1	Foggy	Khammam Bypass Road
6009	12/17/2011	11:04:00 AM	Mysuru Road	64	Bike	Minor	0	1	Rainy	Mysuru Road Road
6010	04/19/2011	7:21:00 PM	Mysuru Road	64	Truck	Fatal	2	1	Foggy	Mysuru Road Road

Contributer – Kasu Gowthami Reddy

Role – Database Developer

Technologies used:

MySQL,
SQL

Implementation:

1.Designed the database structure for road accident records.
2. Created tables to store accident statistics.
3. Worked on SQL queries to retrieve data based on the selected year.

Outcomes:

Developed a structured database to store and retrieve road accident information efficiently.

Backend development using flask (Charan)

app.py

app.py > dashboard

```python
from flask import Flask, render_template, request, jsonify
import mysql.connector

app = Flask(__name__)

def get_db_connection():
    return mysql.connector.connect(
        host="localhost",
        user="root",
        password="[redacted]",
        database="road_accident_db"
    )


@app.route("/")
def home():
    return render_template("index.html")


@app.route("/api/dashboard")
def dashboard():
    conn = None
    cursor = None

    try:
        year = int(request.args.get("year", ""))

        if year < 2011 or year > 2026:
            return jsonify({
                "error": "Please enter a year between 2011 and 2026."
            }), 400

        conn = get_db_connection()
        cursor = conn.cursor(dictionary=True)
```

Figure: Flask Backend Code for Routes and Database Connection

Problems Output Debug Console Terminal Ports

```
PS C:\RoadAccidentProject> python app.py
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 873-585-417
```

Figure: Flask Application Running on http://127.0.0.1:5000

Contributer – Anneboyina Venkata Sai Charan

Role – Backend Developer

Technologies used:

Python,
Flask,
MySQL

Implementation:

1. Developed backend functionality using Flask.
2. Connected the web application with the database.
3. Worked on retrieving accident statistics and sending data to the dashboard.

Outcome:

Connected the frontend and database to support dynamic display of road accident statistics.

Frontend Development (Harsha Bindu)

Dashboard webpage and frontend code

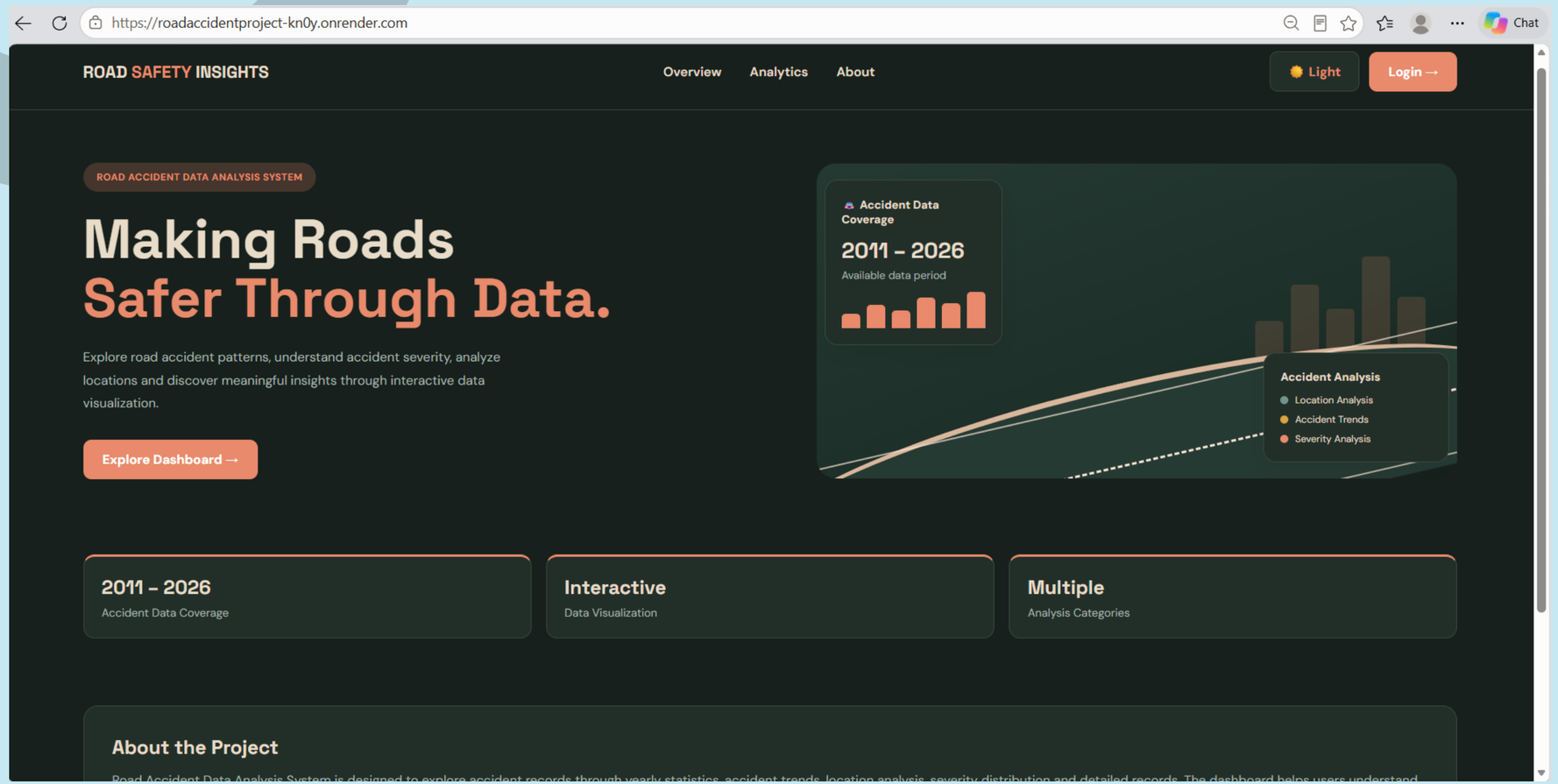


Figure: Landing Page of the Road Accident Data Analysis System

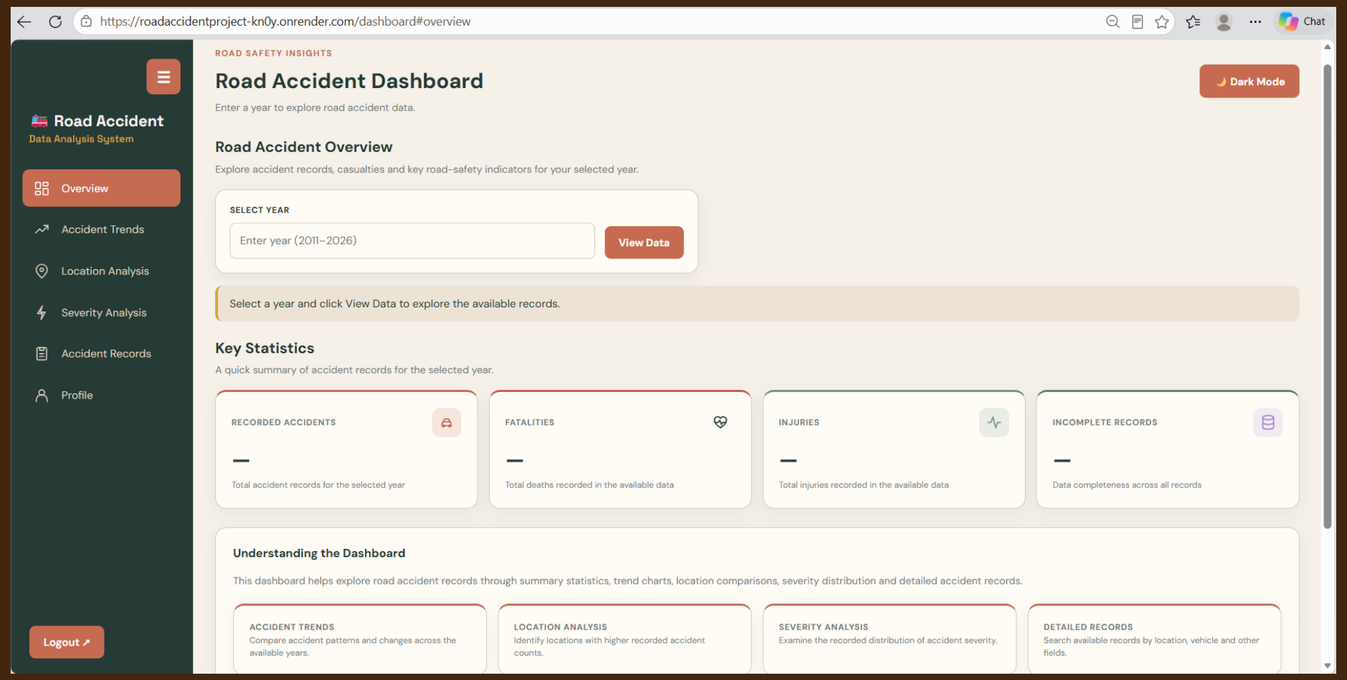


Figure: Overview Page which Displays Data for the Selected Year

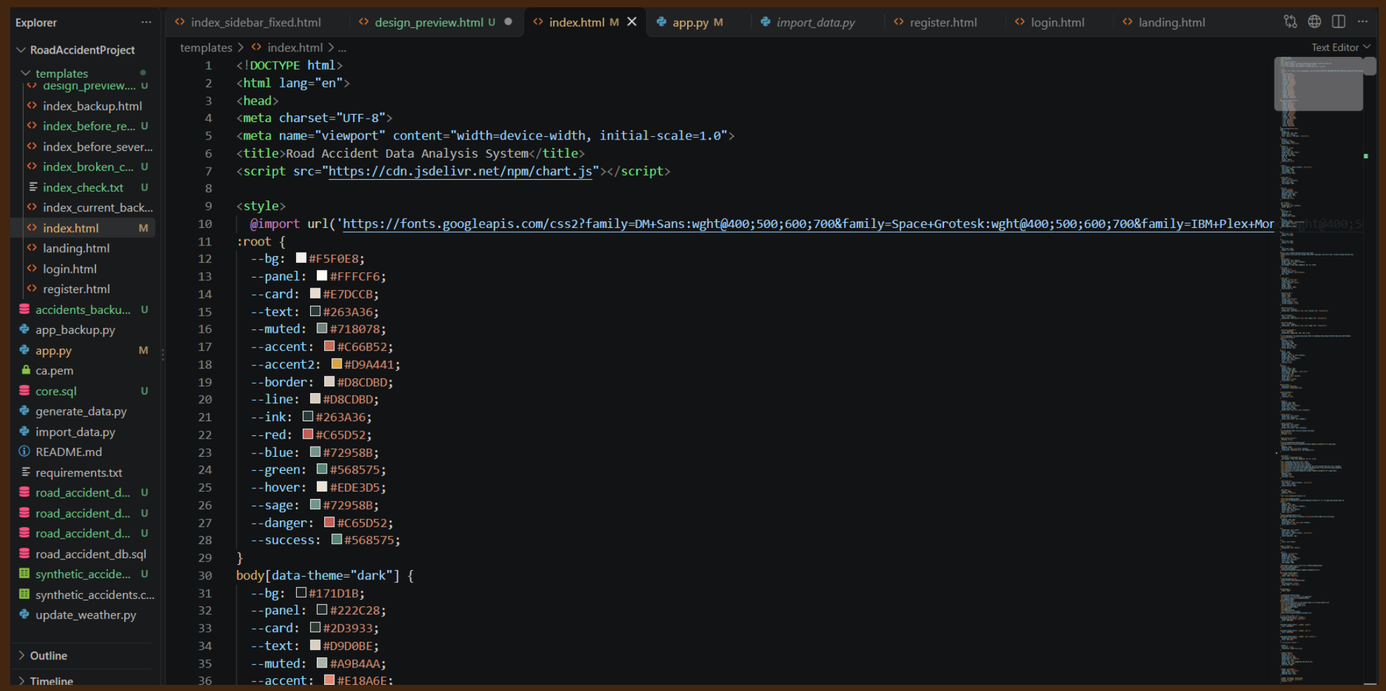


Figure: HTML Code for the Dashboard Frontend

Contributer – Kalikiri Harsha Bindu

Role – Frontend developer



Technologies used:

HTML
CSS

Implementation:

1. Designed the road accident dashboard interface.
2. Created input fields for selecting accident data by year.
3. Developed sections to display accident statistics and charts.

Outcome:

Created a user-friendly dashboard for viewing and understanding road accident statistics.

THANK YOU!

